

Executive Summary

The Indian Real Estate Price Prediction System is a machine learning based application developed to estimate residential property prices across Indian cities. The system combines data preprocessing, feature engineering, machine learning model training, API development, and web-based visualization into a complete end-to-end solution.

The primary objective of the project is to help buyers, sellers, and real-estate professionals obtain quick and data-driven property valuations. The application uses historical property data and key features such as city, neighborhood, property type, number of bedrooms, number of bathrooms, and property size to generate predictions.

The solution consists of an XGBoost machine learning model, a FastAPI backend for prediction services, and a Streamlit frontend for user interaction. The project demonstrates practical implementation of machine learning deployment and serves as a portfolio-ready capstone project.